American Journal of Transplantation 2004; 4: 1662–1668
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doi: 10.1111/j.1600-6143.2004.00573.x

Kidney Transplantation Halts Cardiovascular Disease Progression in Patients with End-Stage Renal Disease

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Morbidity and mortality from cardiovascular disease have a devastating impact on patients with chronic kidney disease (CKD) and end-stage renal disease. Renal function decline in itself is thought to be a strong risk factor for cardiovascular disease (CVD). In this study, we investigated the hypothesis that the elevated CV mortality in kidney transplant patients is due to the preexisting CVD burden and that restoring renal function by a kidney transplant might over time lower the risk for CVD. We analyzed 60 141 first-kidney-transplant patients registered in the USRDS from 1995 to 2000 for the primary endpoint of cardiac death by transplant vintage and compared these rates to all 66813 adult kidney wait listed patients by wait listing vintage, covering the same time period. The CVD rates peaked during the first 3 months following transplantation and decreased subsequently by transplant vintage when censoring for transplant loss. This trend could be shown in living and deceased donor transplants and even in patients with end-stage renal disease secondary to diabetes. In contrast, the CVD rates on the transplant waiting list increased sharply and progressively by wait listing vintage. Despite the many mechanisms that may be in play, the enduring theme underlying rapid progression of atherosclerosis and cardiovascular disease in renal failure is the loss of renal function. The data presented in this paper thus suggest that the development or progression of these lesions could be ameliorated by restoring renal function with a transplant.

Key words: Cardiovascular death, dialysis, kidney transplantation, renal function, survival benefit

Received 28 April 2004, revised and accepted for publication 4 June 2004

Introduction

It is well established that patients with chronic renal failure suffer higher rates of cardiovascular morbidity and mortality than the general population. As early as in 1974, Lindner et al. recognized that patients on dialysis had dramatically rapid progression of atherosclerosis (1). In fact, more recent data have confirmed that cardiovascular mortality increases by approximately 10 folds in patients with end-stage renal disease, even after accounting for effects of patient age, sex, race, and the presence of diabetes (2–4). Ischemic heart disease and congestive heart failure are each present in about 50% of the patients beginning dialysis (2), and the case fatality rate for each of these diseases in patients with end-stage renal disease (ESRD) is extremely high (3,5).

A more alarming finding is that total absence of renal function is not the only factor associated with increased cardiovascular mortality in chronic renal failure. Shulman et al. (1989) demonstrated that an elevated serum creatinine level was one of the strongest predictors for mortality (6). More recently, even mild decreases in renal function have been correlated to the risk for cardiovascular death (7).

Renal transplantation confers a significant survival advantage over maintenance dialysis over and beyond what one expects based on selection bias involved in the transplant process (8). Cardiovascular death remains one of the predominant causes of graft loss after kidney transplantation (9,10) and also the major cause of fatality on dialysis (11). It is likely that a large part of this survival advantage is connected to decreases in both progression of cardiovascular disease and cardiovascular mortality following a kidney transplant.

This survival advantage conferred by successful renal transplantation is evident despite the many potentially deleterious effects of renal transplantation on many of the cardiovascular risk factors, which are then superimposed on this high-risk population of patients who have had chronic renal failure or dialysis for varying periods before ultimately receiving a kidney transplant. The immunosuppressive regimens *per se* can cause or worsen hyperlipidemia (12), hypertension (13), anemia (14), and diabetes

(15), each of which may in turn hasten the progression of cardiovascular disease (10,12,16–19).

The association between renal failure and cardiovascular disease can to some extent be explained by the presence of predisposing factors shared by the two illnesses, such as hypertension, lipid abnormalities, and cigarette smoking (20). On the other hand, numerous studies demonstrate that even mild renal insufficiency could predispose to the development and progression of cardiovascular disease (3,21–24). The importance of these findings is enhanced by the knowledge that certain treatment options slow this process, at least partially (25).

While all the studies cited thus far indicate an effect of renal function independent of potentially confounding variables like smoking and hypertension, none of these studies were able to exclude risk factors predisposing to both renal and cardiac disease constituting the link between decreasing renal function and development (and/or progression) of cardiac disease. To date, the only unequivocal evidence for decreased renal function as an independent mediator of accelerated atherosclerosis comes from the studies on animal models (26).

Renal transplantation, on the other hand, provides an excellent human model to investigate whether or not diminished renal function is a cardiovascular risk factor independent of other risk factors common to both disease processes because it restores (albeit to a variable degree) lost renal function. Furthermore, the level of function of a transplanted kidney at 1-year post-transplantion is mostly an expression of the quality of the organ and the immunological factors that could have had a damaging impact, and is largely independent of recipient factors predisposing the patient to cardiac complications (27). We have shown in a previous study that impaired renal function at 1 year after a kidney transplant is a strong risk factor for cardiovascular death (27), an indication that good renal function *per se* probably has some intrinsic effects protecting the recipient from development or progression of cardiovascular disease.

To test our hypothesis that improvement in renal function conferred by renal transplantation halts or even reverses the progression of cardiovascular disease, we analyzed data from the USRDS registry for the occurrence of cardiovascular death by dialysis versus transplant vintage.

Methods

We analyzed data pertaining to 60141 first solitary adult (age $\geq$ 18) kidney-transplant recipients between 1995 and 2000 from the USRDS database to examine the cardiovascular (CV) and overall death rates of this cohort. In addition, we performed a complementary analysis of all 66813 adult kidney wait-listed patients covering the same time period from the USRDS database in order to calculate similar rates. We calculated rates separated by transplant type (deceased donor and live donor) and conducted a subanalysis of recipients with diabetes and hypertension as a primary diagnosis. Diabetic patients included both type-I and type-II diabetes, while hypertensive patients were classified with a primary diagnosis of malignant hypertension or hypertensive nephrosclerosis. The primary outcomes of our study were patient death and cardiovascular death (as primary cause). Wait-list patients were censored at the time of transplant. We also analyzed outcomes of patient death and CV death censored at the time of graft loss. Deaths classified as CV derived from primary causes of death listed in the registry as any of the following: pericarditis, myocardial infarction, cardiac (other), myocardial infarction (acute), atherosclerotic heart disease, cardiomyopathy, cardiac arrhythmia, cardiac arrest and valvular heart disease. Patients with relevant death dates, but missing or uncoded death types, were considered as deaths by non-CV causes. As a result of this assumption, the incidence of CV deaths is notably underestimated in this analysis as these death types represented a significant percentage of all deaths.

Transplant vintage, defined as the time post-transplant until the applicable outcome or last follow-up time, was categorized as 0–3, 3–6, 6–12, 12–24, 24–36, 36–48, 48–60 months and >60 months. In a similar fashion, wait-list vintage was calculated using the same durations with first wait-list time as the initial vintage start date (e.g., an event occurring within 3 months post-listing would be treated equivalently for a candidate listed in June 1995 as one listed in November 1998). Crude death and CV death rates were calculated by tabulating the number of applicable events within the vintage categories divided by the amount of follow-up time (in years) for the cohort, and then dividing by 1000 to depict the rate by 1000 patient years. Follow-up time for patients was recorded as complete for the applicable cases in which their time was beyond the vintage (e.g., a patient with 5 months of follow-up contributed 0.25 patient years (3 months) to the 0–3 month vintage), and fractional contributions were utilized for those cases that ended within the vintage relative to the amount of time they were eligible for the events. We also examined death rates for the transplant cohort by the length of time on dialysis prior to transplant. Time on dialysis was categorized into preemptive, 0–6, 6–12, 12–24, 24–36, 36–48, 48–60 months and 60+ months.

Death rates were additionally adjusted by recipient age for both the transplant and the wait-list cohorts. To calculate these rates, we categorized recipient age (applicable to vintage of their respective follow-up) to 18–24, 25–34, 35–44, 45–54, 55–64 years, and 65+ years. We tabulated events and follow-up time within each age category (as described previously) by vintage. We then directly adjusted the rates by weighting them by the relative frequency of the age categories within the cohort (e.g., 0–3 month vintage death rates were calculated by the rates in 18–24 years olds multiplied by the frequency of this age group added to the rates in the 25–34 years old group, etc.). For results separated by transplant type, the rates were weighted by the respective age relative frequencies, separately for living and deceased donor populations.

We additionally analyzed the time to cardiovascular death by the duration of pre-transplant dialysis using Kaplan–Meier plots and tested for significant association of the strata of dialysis time with CV death rates with the Log-Rank test.

The last follow-up time in the entire cohort was July 21st, 2001. Analyses were conducted with SAS v.9.0 (Cary, NC).

Results

The age-adjusted rates for deceased donor transplants uncensored for graft loss are displayed in Figure 1. As evidenced by this graph, the overall and CV death rates in

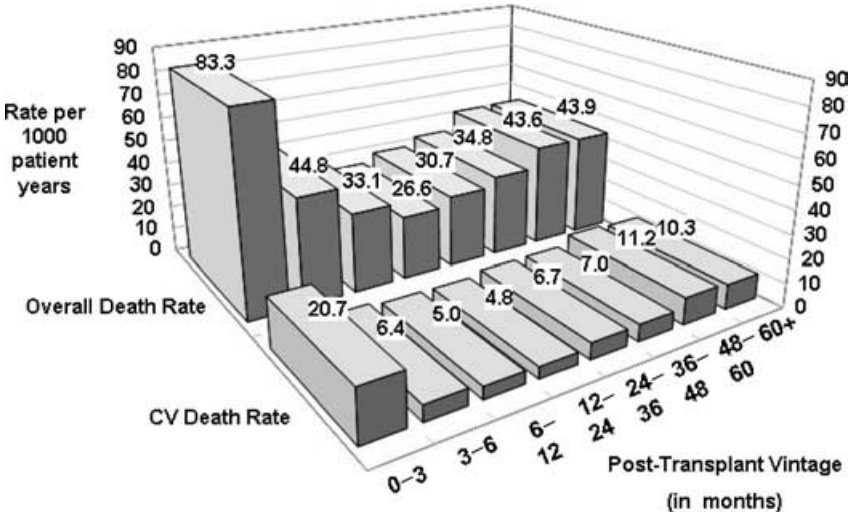


Figure 1: Overall and cardiovascular death rates by post-transplant vintage for deceased donor transplant recipients 1995–2000.

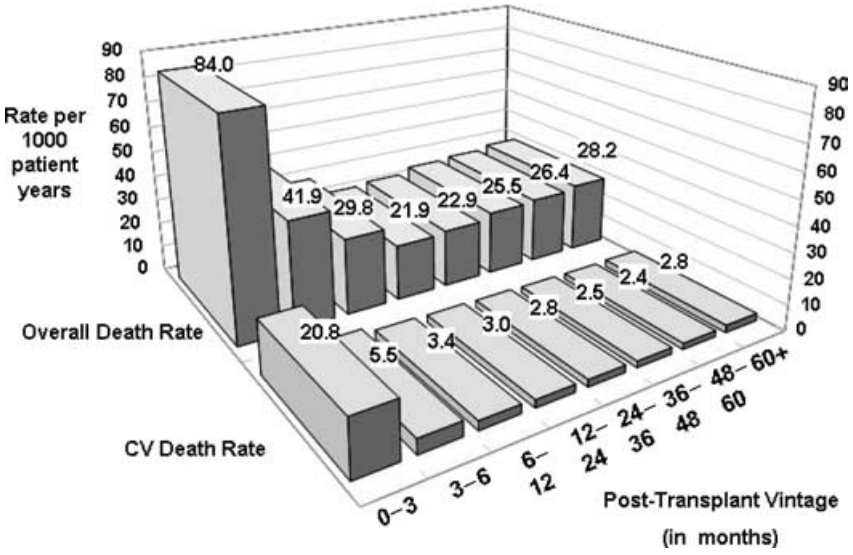


Figure 2: Overall and cardiovascular death rates by post-transplant vintage for deceased donor transplant recipients 1995–2000 (censored at graft loss).

the immediate post-transplant period (0–3 months) were highest, indicating that this is the period of maximum risk. The trajectory of risk for both death and CV death declined initially and subsequently ascended gradually in the later post-transplant period. The ratio of CV to overall death rates almost halved from the initial 0–3 month period to the 3–6 month period, but then steadily increased along with the magnitude of the individual rates. The unadjusted crude rates were very similar in magnitude and trend for each of these outcomes.

We repeated the analysis for deceased donor recipients adjusting the model by censoring follow-up at the time of graft loss. These data are shown in Figure 2. By applying the censoring condition, the CV rates still peak at the initial 3-month period, but remain relatively flat throughout the remainder of the follow-up period, which we studied. The CV to overall death rate dropped from the 0–3 to the 3–6 month follow-up vintages and maintained a similar ratio throughout the remainder of the study period. Unadjusted rates mirrored this trend very closely. We then examined these trends for living transplant recipients in this era. The results, as displayed in Figure 3 (for uncensored follow-up) and Figure 4 (for the censored analysis), indicate the expected lower magnitudes of death and CV death rates for the living transplant recipients. However, the trend between the censored and the uncensored rates by vintage in the living transplant recipient cohort was very similar to the deceased donor rates. The most frequent causes of CV death in this study were cardiac arrest (45.1%), myocardial infarction (31.1%), cardiac arrhythmia (12.8%), atherosclerotic heart disease (5%) and cardiomyopathy (4.1%). We also examined the rates of death and CV death in the wait-list population over the same era. The adjusted rates for the entire wait-list population (censored at the time of transplant) can be viewed in Figure 5. For this cohort, the rates appear to increase steadily by dialysis vintage for both overall and CV death. Of note, however, while the sample was

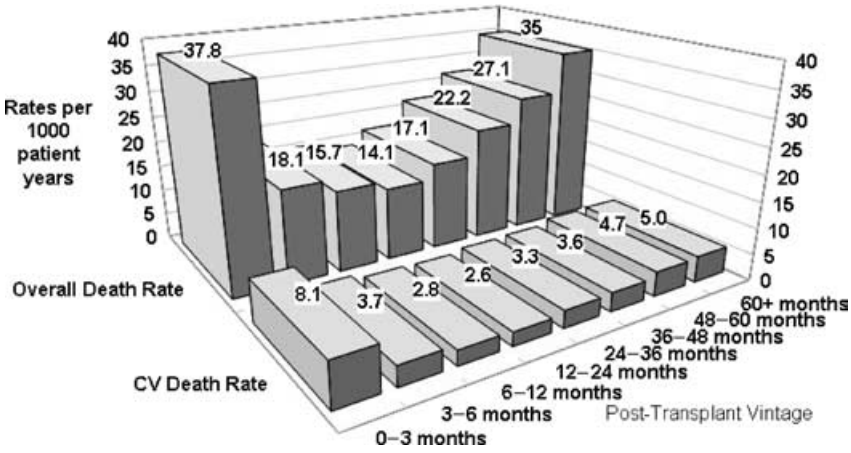


Figure 3: Overall and cardiovascular death rates by post-transplant vintage for living transplant recipients 1995–2000.

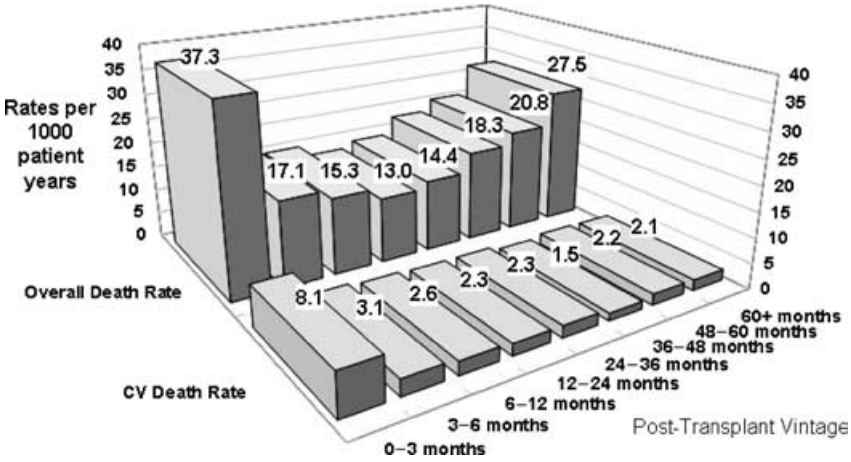


Figure 4: Overall and cardiovascular death rates by post-transplant vintage for living transplant recipients 1995–2000 (censored at graft loss).

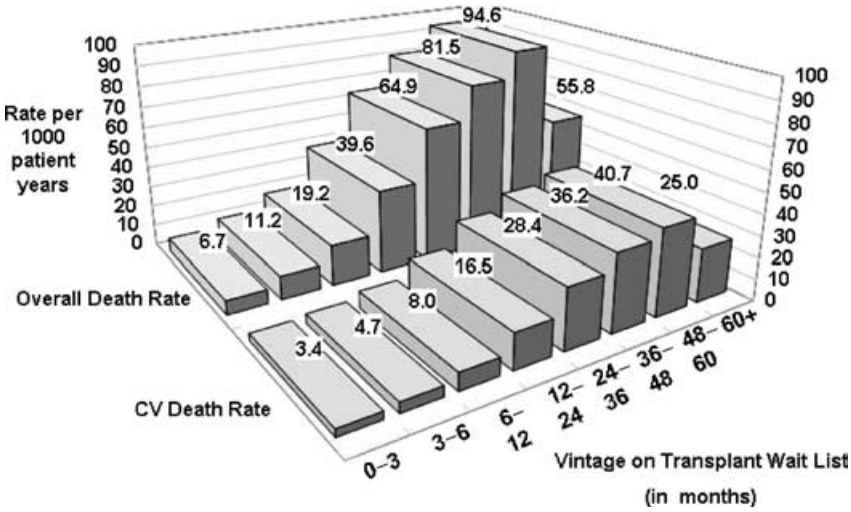


Figure 5: Overall and cardiovascular death rates by wait-list vintage 1995–2000 (censored at transplant).

relatively small for past 60 months, rates declined in this period relative to the pattern preceding it. The ratio of CV death to overall death remained relatively stable for the entire follow-up period ranging between 40 and 50% of all deaths. The ratio of CV deaths was substantially higher in the wait-list population relative to the transplant-recipient population censored for graft loss when comparing by vintage.

Repeating the analysis in a particularly high-risk population, we examined the CV and overall death rates in recipients with diabetes as a primary diagnosis. We also separately

calculated CV and overall death rates in recipients with a primary diagnosis of hypertension. Results for deceased donor recipients with both diagnoses censored at graft loss are shown in Tables 1 and 2, respectively. While the magnitude of the death rates in the diabetic population was, as expected, higher than the overall population, when censoring for graft loss, the CV death rates stabilized by post-transplant vintage. A similar trend was noted for the hypertensive transplant recipients with their death rate magnitudes similar to the overall population.

Kaplan–Meier plots of time to CV death for first solitary transplant recipients who received their transplant between 1990 and 2000 are displayed in Figure 6. The Log-Rank test was highly significant ($p < 0.0001$) with an increasing hazard associated with more extended time on dialysis. This relationship is further illustrated in Table 3, depicting the increasing CV and overall death rates in the 1995–2000 transplants by time on pre-transplant dialysis.

Discussion

Our study demonstrates a progressive decrease in cardiovascular death rates by renal transplant vintage (Figure 2). This trend is even more meaningful when contrasted with the opposite trend by dialysis vintage (Figure 5). Even though cardiovascular death rates were expectedly higher during the early post-operative period after surgery, at 3 months post-surgery the cardiovascular death rate dropped significantly to lower than that for patients remaining on the waiting list. The early cardiovascular death rates after renal transplantation are in fact modest, when compared to cardiovascular death rates of patients for 2–3 years on the dialysis waiting list, considering this represents the average waiting time of the cohort studied.

Table 1: Age-adjusted CV and overall death rates for deceased donor transplants by post-transplant vintage for recipients with diabetes as primary diagnosis (censored at graft loss)

Transplant vintage (months)	Total patients eligible	CV death rate*	Overall death rate*
0–3	12409	36.4	119.1
3–6	11524	9.1	55.3
6–12	11214	5.8	38.8
12–24	10154	5.6	35.1
24–36	8219	5.1	34.4
36–48	5906	4.1	41.3
48–60	3810	3.8	37.9
60+	2091	4.0	43.8

*Adjusted by age distribution from all transplants for recipients with diabetes as primary diagnosis; rates are per 1000 patient years.

Table 2: Age-adjusted CV and overall death rates for deceased donor transplants by post-transplant vintage for recipients with hypertension as primary diagnosis (censored at graft loss)

Transplant vintage (months)	Total patients eligible	CV death rate*	Overall death rate*
0–3	8894	19.2	88.8
3–6	8254	6.0	41.2
6–12	7995	3.5	39.1
12–24	7003	3.2	26.3
24–36	5373	3.9	31.3
36–48	3814	2.2	29.6
48–60	2391	2.7	38.2
60+	1246	2.4	33.9

*Adjusted by age distribution from all transplants for recipients with hypertension as primary diagnosis; rates are per 1000 patient years.

On the other hand, there seemed to be a late rise in cardiovascular death rates by transplant vintage (Figure 1). This is not too surprising as renal function after transplantation has been shown to be a strong risk factor for cardiovascular death (27), and as renal function gradually declines and graft loss increases with increasing transplant vintage, we expect the cardiac risk of the patients to increase with time (28). Censoring the data for graft loss (i.e., considering only patients with a functioning kidney transplant), however, reveals the cardiovascular death rates remained impressively low even in long-term transplant vintage patients. This even held true in the high cardiovascular risk group of patients with end-stage renal disease secondary to diabetes (Table 1) where, as expected, the cardiovascular death rates were about twice as high as among nondiabetic patients. Similar trends were observed in the other high cardiovascular risk group of patients with end-stage renal disease secondary to hypertension.

Probably the best comparison between dialysis and transplantation for the cumulative risk of cardiac death and cardiovascular disease progression is the comparison between the cardiovascular death rates (displayed in Figure 2) for patients who retain a functioning graft and (Figure 5) patients who remain wait listed on dialysis. Early on, when patients with end-stage renal disease are enrolled on the transplant waiting list, the cardiovascular death rate is low. That is probably due to the rigorous screening processes the patients undergo during the initial evaluation (i.e., wait-listed patients have fewer cardiovascular risk factors and less severe cardiovascular disease). Only 1 year after being on dialysis, the cardiovascular death rates had more than doubled and were significantly higher than 1 year after transplant. The significance of our data resides not in the absolute rates but rather in the trend over time and the persistently low cardiovascular death rates by transplant vintage as long as function of the kidney graft is preserved. This is in striking contradistinction to the steady rise of cardiovascular deaths by dialysis vintage.

Cardiovascular death rates increase by dialysis vintage secondary to accelerated cardiovascular disease progression. Therefore, in addition to more patients dying on the wait list with increasing vintage, those who survive will accrue a higher mortality risk from cardiovascular death because

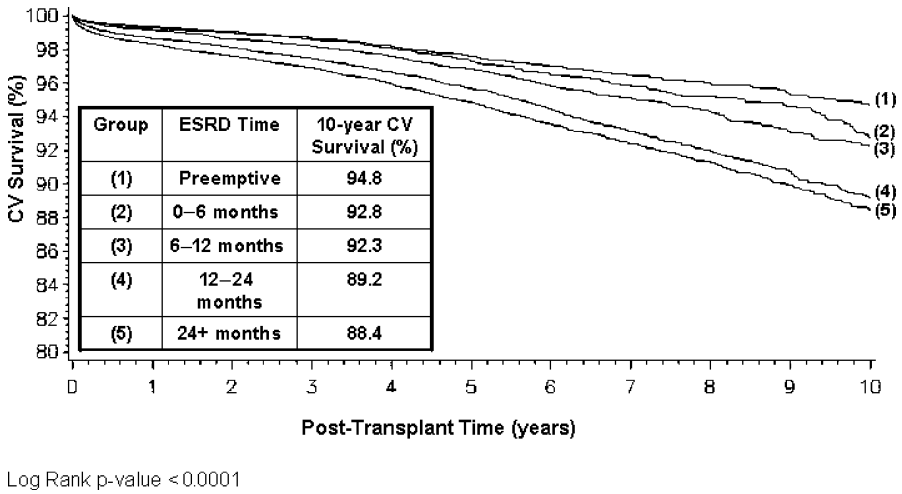


Figure 6: Kaplan–Meier plot of cardiovascular death by pre-transplant dialysis time (transplants 1990–2000).

Table 3: Age-adjusted CV and overall death rates for deceased donor transplants by pre-transplant dialysis time (uncensored at graft loss)

	1-year rates		3-year rates	
Pre-transplant dialysis time	CV death rate*	Overall death rate*	CV death rate*	Overall death rate*
Preemptive	7.5	54.0	4.9	35.6
0–6 months	7.2	40.1	7.4	33.2
6–12 months	10.3	46.7	8.0	38.4
12–24 months	10.5	54.0	8.2	42.3
24–36 months	13.0	62.5	9.4	49.0
36–48 months	14.6	68.1	10.6	52.6
48–60 months	12.8	67.6	10.3	51.9
60+ months	17.1	86.8	12.5	61.8

*Adjusted by age distribution from all deceased donor transplants; rates displayed per 1000 patient years.

of advanced disease. This explains the strong impact of waiting time on dialysis as a risk factor for cardiovascular death after transplantation (Figure 6).

As cardiovascular death takes the biggest mortality toll in patients on maintenance dialysis (11), a large component of the survival advantage of kidney transplantation previously shown (8) is probably due to a long-term reduction of the cardiovascular disease progression and mortality.

Whenever transplantation is compared to the waiting list, there is a possible intrinsic bias against patients on the waiting list who are not transplanted because they are medically not suitable for transplantation but are not officially removed from the list. Even though this bias could be increasing the measured effect, it is unlikely to explain the whole phenomenon. A strong argument against this being a significant factor is that only a minor proportion of the patients on the deceased donor waiting list actually receive a transplant, while the majority of very suitable patients remain waiting because of the scarce organ availability.

Substantial research evidence have accumulated to explain the pathogenesis of cardiovascular disease in chronic renal failure and end-stage renal disease (11). Considering the progressive arteriopathy accompanied by inflammation (29–32) and calcification (30,33) in patients on dialysis, and the multitude of pathogenetic mechanisms underlying the vasculopathy, the magnitude of difference in cardiovascular death in transplant patients with restored renal function is no surprise. Despite the many mechanisms that may contribute, the enduring theme underlying rapid progression of atherosclerosis and cardiovascular disease in renal failure is the loss of renal function. The data presented in this paper thus suggest that the development or progression of these lesions (as inferred from cardiovascular mortality) could be ameliorated by restoring renal function with a transplant.

Clearly a functioning kidney transplant confers significant lasting protection from cardiovascular death, probably by halting or possibly even reversing cardiovascular disease progression in ESRD patients.

Acknowledgments

The data reported here were supplied by the U.S. Scientific Renal Transplant Registry (SRTR). The interpretation and reporting of these data are the responsibility of the authors and in no way represent an official policy or interpretation of the U.S. Government. Part of these data are going to be presented at the annual meeting of the American Society of Transplantation in Boston in 2004. We would like to express our appreciation for Suzanne C. Johnson who has helped with the editing and review of the paper.

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